

# Object Oriented Programming Lab

## Lab 13

### Instructions

Work on this lab individually. You can use your books, notes, handouts etc but you are not allowed to borrow anything from your peer student.

Write down the lab number, task number, your name and roll number at the start of each task code using comments.

For Example

```

/*****
Lab:    xx
Task:   yy
Developer:
        Roll #: XXXYYYZ000
        Name:  XYZ
*****/

```

### What you have to do

Program the following tasks in C++, compile and execute them.

### Task

Create base class **Device** with following attribute and functions:

- string deviceID
- Constructor prints: "Device constructor"
- void showID()

Create **SmartLight : public Device**

- Constructor prints "SmartLight constructor"
- void adjustBrightness()

Create **SmartFan : public Device**

- Constructor prints "SmartFan constructor"
- void adjustSpeed()

Create **SmartCombo : public SmartLight, public SmartFan**

- Constructor prints "SmartCombo constructor"

In main():

Part 1:

```
SmartCombo sc;
sc.showID();    // should cause ambiguity
```

Part 2:

Fix ambiguity using virtual inheritance

Part 3:

Add virtual void status() in **Device** class, override in child classes, and make polymorphic calls. That is,

```
Device* d = new SmartCombo();
d->status();
```

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